Where the compile-time guarantee stops

Rust eliminates data races, not race conditions; aliasing errors, not lock-ordering errors

INSIDE THE GUARANTEE

Rejected by the compiler, on every build, covering every execution of the program

Lifetime and aliasing

enforced by ownership and borrowing

- Use-after-free
- Double-free
- Iterator invalidation

Unsynchronised access to a location

enforced by Send/Sync — these are *data races*

- Data race
- Unguarded shared mutation

In C++ both compile; only ThreadSanitizer reports them.

No test coverage required for the guarantee to hold

OUTSIDE THE GUARANTEE

Compiles in safe Rust and misbehaves, exactly as its C++ counterpart does

Order of lock acquisition

not prevented by either compiler

- Lock-order deadlock — hangs

The borrow checker tracks a guard's lifetime perfectly; that says nothing about the order threads take locks.

Atomicity of a composed operation

not prevented either — these are *race conditions*

- Lost update — silently wrong value

Every access is correctly synchronised, so there is no race to find: no tool in either toolchain reports it.

Fails quietly: a hang or a corrupted counter, not a crash

the boundary

The phrase "thread safety" spans both columns; Rust's guarantee covers only the left one.

It retires one category of concurrency defect entirely and leaves the other in place — and the one left in place fails silently.